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User perception based label layout for efficient target localization in virtual environment

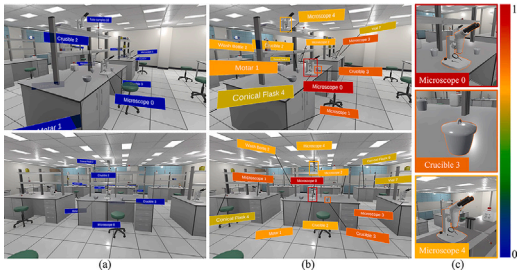
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HIGHLIGHTS

- We introduce a new concept of object interest score that captures a user's interest in objects and propose an algorithm for updating the object interest score in real-time.
- We propose a data-driven approach for modelling the mapping of bootstrap label attributes to user-perceived time to measure user reaction time to labels in the field of view.
- We propose a dynamic guidance label layout method driven by user perception force.
- We introduce a new metric, "label perceived time cost," for measuring the user's perceived cost of the current label layout.

GRAPHICAL ABSTRACT



The comparison of two user views (rows 1 and 2) with the traditional dynamic label layout method (a) with the label filtering method based on the distance from the center of the view proposed by McNamara et al. (2019) and with our label layout method (b). Objects with different interest scores that may be of interest to the user are shown in (c).

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ABSTRACT

Virtual Reality (VR) technology provides users with an immersive experience in virtual environments. In target localization scenarios, labels are essential for guiding users to quickly identify and locate targets, thereby enhancing localization efficiency. However, the increasing complexity and volume of information in virtual environments often leads to label information overload and label visual clutter. To address these challenges, we propose a user perception-based label layout method to improve target localization efficiency. This method introduces the concept of object interest score, which dynamically quantifies user interest in scene objects over time and is used for label filtering. Additionally, a mapping function from label attributes to user-perceived time is constructed, allowing the estimation of how quickly a user can perceive a label. By leveraging both the object interest score and user-perceived time, we develop a user perception force-driven label layout optimization method. To evaluate the layout's effectiveness, we propose a new metric, label perceived time cost, which measures the effort required for users to perceive the labels of objects of interest. User studies show that the proposed method significantly improves target localization efficiency, reduces task load, enhances usability, and minimizes sickness compared to traditional dynamic force-based label layouts.

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1. Introduction

Recently, virtual reality applications have deepened in several fields, including manufacturing and healthcare. Target localization is widely used in virtual assembly, repair, and training. Some work improves target localization efficiency by using visual cues such as salient colors Erickson et al. (2022) or highlighting target outlines Horst et al. (2019), and reducing occlusions Yu et al. (2019). However, when users need to find a specific object among many similar-looking candidates, visual saliency cues such as color or shape may not be sufficient to distinguish objects, and they must specify a search behavior based on specific semantic information. The label-guided localization can accelerate users' localization of targets in virtual environments through intuitive visual cues, which is particularly effective when users need to quickly find the desired target from a large number of objects. Researchers have carried out a lot of work on label layout methods in VR/AR. Aiming at the problem of overlapping labels and lead crossing under some viewpoints for labels, Hartmann et al. (2004) proposed a label layout method based on the particle spring model to eliminate overlapping and crossing as much as possible. A large number of objects in a complex scene will lead to too many labels, to address this problem, McNamara et al. (2019) selected labels according to the user's gaze area, thus reducing the number of labels displayed. Madsen et al. (2016) showed in a detailed user study on labelings in augmented reality that the participants performed better in the conducted tasks when the labels were directly placed in the 3D space of the object instead of the 2D space. Therefore, we optimize labels in 3D space to maintain depth perception.

However, utilizing label guidance to quickly locate a target in a complex virtual scene containing many objects still poses some challenges for label layout. The first challenge is to find objects that are likely to be of interest to the user. When searching for a target, users will naturally look at objects that are similar in shape and placement to the target, so these objects are likely to be of interest to the user. Helping users filter objects of interest can effectively avoid the problem of label information overload. The second challenge is to determine the 3D location of these labels so that users can find them quickly. When a user wears a VR HMD to observe a virtual scene, the user usually reacts faster to the content in the center of view than in other areas. Biological literature (Schiefer et al., 2001) also mentions that human visual reaction time is positively correlated with the eccentricity angle of the content in the field of view. Laying out labels in locations that users perceive more quickly can improve their search efficiency. The third challenge is how to rationally lay out the labels in these 3D locations. Although the existing particle spring model can move the labels around the original position according to the user's viewpoint, thus reducing label overlap and wire crossing, it cannot push the labels of invisible objects that may be of interest to the user to the visible position or reasonably lay out the labels to a position that can be quickly perceived by the user.

In this paper, we propose a user perception based label layout method for target localization. This method improves the efficiency of label-guided target localization by analyzing the user's interest scores in objects during the search process and placing labels for objects with a higher interest in 3D locations with a shorter user perception time. First, we define object interest scores to characterize the user's interest in the objects in the scene over time. Second, we propose a data-driven fitting function method to create a mapping from label attributes to user-perceived time, quantifying how quickly a label is recognized and processed by a user. This mapping function can be used to estimate the user's perceived time for the labels of objects that the user may be interested in. Finally, we calculate the user perception force based on the object interest score and user-perceived label time function and propose a dynamic guided label layout optimization driven by user perception force to dynamically adjust the label layout. In addition, we introduce a new metric, label perceived time cost, to measure the cost required for users to perceive the labels of their objects of interest based on the current labeling layout. We designed a user study to evaluate our method. Target localization with our label layout significantly increases efficiency, as well as reduces task load, and improves usability, while also reducing sickness, compared to localizing targets guided by traditional dynamic label layouts.

Fig. 1 shows the comparison of two user views (rows 1 and 2) with the traditional dynamic label layout method (a) with the label filtering method based on the distance from the center of the view proposed by McNamara et al. (2019) and with our label layout method (b). In both views, the user is looking at $Microscope0$ highlighted with a red square. Using the traditional dynamic label layout method (a), only the n closest labels to the center of the user's field of view are displayed, and the labels are usually located in the neighboring regions of the corresponding objects. The particle spring system reduces label overlap and lead crossing but may result in label occlusion because the labeled object is located in the center of the field of view range and the depth varies. Using our label layout method (b), we use the legend color on the right for label visualization; the redder the label color, the more interested the user is in the object identified by the label, and we visualize labels only for the top n objects that are of most interest to the user, with labels for objects of higher interest located at more easily perceived 3D locations. Based on the $microscope0$ that the user is currently gazing at, our method estimates that the user is interested in objects with a certain degree of similarity, e.g., $microscope3$, $crucible3$, $microscope1$ because the

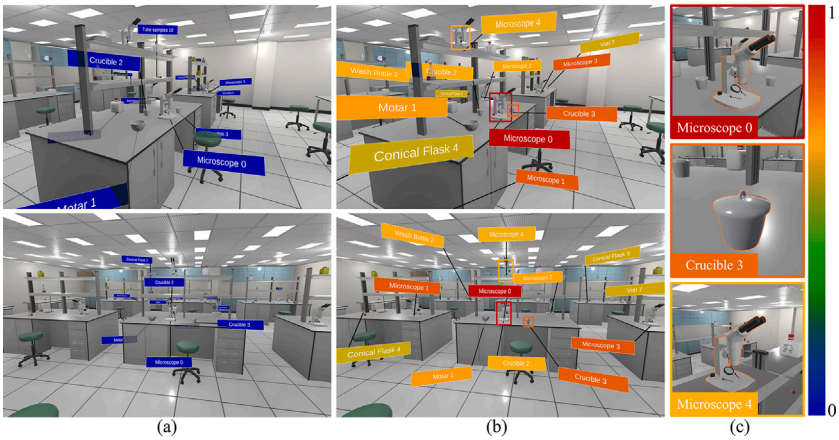


Fig. 1. The comparison of two user views (rows 1 and 2) with the traditional dynamic label layout method (a) with the label filtering method based on the distance from the center of the view proposed by McNamara et al. (2019) and with our label layout method (b). Objects with different interest scores that may be of interest to the user are shown in (c).

shapes of *microscope*3, *microscope*1, and *microscope*0 are basically the same, the environments of *microscope*3, *crucible*3, and *microscope*0 are more similar, while *microscope*1 and *microscope*0 are less similar, and *microscope*4 has a completely different environment than *microscope*0. (c) shows a magnified view of *microscope*0, *crucible*3, *microscope*4 and their surroundings.

In summary, our main contributions are as follows:

- We introduce a new concept of object interest score that captures a user's interest in objects and propose an algorithm for updating the object interest score in real-time.
- We propose a data-driven approach for modelling the mapping of bootstrap label attributes to user-perceived time to measure user reaction time to labels in the field of view.
- We propose a dynamic guidance label layout method driven by user perception force.
- We introduce a new metric, "label perceived time cost," for measuring the user's perceived cost of the current label layout.

2. Related work

In this section, we first introduce some works on target localization using visual cues, and then we review previous methods for 3D label layout and label filtering in virtual and mixed scenes.

2.1. Target localization

In a virtual environment, various forms of signals can be used to provide users with clues for target positioning Baldwin et al. (2012); Trepkowski et al. (2021), such as visual signals, auditory signals, tactile signals, etc. For example, Wang et al. focused on investigating and designing visual and haptic guidance systems to augment users' performance Wang et al. (2024). They found that haptic guidance can enhance both reaction time and user experience in dual-task environments. Bernal et al. explored the impact of audio-visual offsets and rotation velocities on users' directional localization acuity for various viewing modes Bernal-Berdun et al. (2024).

Among them, visual clues are a very important form of signal and play a dominant role in target positioning tasks Tobisková et al. (2022). Visual cues play an important role in tasks Stefanucci et al. (2022); Ahn et al. (2021); Dozio et al. (2022) such as navigation, object perception, and target localization. Stefanucci et al. implemented a variety of navigation cues known to facilitate target search and spatial knowledge acquisition in an urban virtual environment Stefanucci et al. (2022). Ahn et al. explored the influence of visual cues on users' perception of distance in virtual scenes, which affects their behavior and interactions Ahn et al. (2021). Horst et al. studied six existing 3D model-based visual highlighting methods for VR scene settings and explored their applicability in 360° video VR scenes Horst et al. (2019). While these attentional visual cues (such as highlighting, coloring, or minimizing occlusion) help draw attention to potential targets, they often lack the semantic specificity required in complex virtual environments, where users need to distinguish between many similar objects. In contrast, labels provide explicit and interpretable information (such as object names), enabling precise target identification. Therefore, our work focuses on enhancing label-based guidance, which remains a necessary and widely adopted strategy for VR target localization.

2.2. 3D label layout

In this section, we mainly discuss previous methods for 3D label layout. For an in-depth understanding of 2D label layouts, we recommend the readers refer to the review by Bekos et al. (2019). 3D label layout plays a key role in enhancing users' understanding of 3D objects. Hartmann et al. (2004) initially applied artificial potential field theory to robotic contour label layouts, innovating optimal label placement by simulating forces between labels and targets. Ali et al. (2005) expanded this concept to 3D visualization by positioning labels along the convex envelope boundaries of objects, introducing methods to prevent lead crossings. Götzelmann et al. (2006) utilized distance fields and ID buffers for label guidance, maintaining layout consistency through proxies in 3D visualization. Stein and Décoret (2008) transformed the 3D layout challenge into a screen-space problem by setting anchor points on 3D objects and projecting them onto a 2D plane. Nöllenburg et al. (2010) examined how map scaling and panning affect label placement, developing an adaptive algorithm. Čmolík and Bittner (2010) identified visible regions and nearest contour points for label placement using greedy algorithms and fuzzy logic. Čmolík and Bittner (2018) and Čmolík et al. (2020) introduced methods for labeling in semi-transparent 3D views and developed a hybrid 2D/3D labeling algorithm to minimize label-guide line overlaps, respectively.

Tatzgern et al. (2014) used 3D geometric constraints: polar constraints and planar constraints to manage the placement of labels in space, allowing the labels to remain temporally coherent as they moved. Madsen et al. (2016) investigated the temporal strategy of AR label layout, using delayed updates when the user's viewpoint exceeds a certain angle or distance to stabilize the label update. It also compares two temporal layout coherence strategies, dynamic labels: delayed update strategy and slow update strategy, and finds that the delayed update strategy is able to perform the task better. They also showed in a detailed user study on labelings in augmented reality that the participants performed better in the conducted tasks when the labels were directly placed in the 3D space of the object instead of the 2D space. Zhou et al. (2021) developed a partially ordered concentric circle layout to simplify label navigation and minimize confusion. Based on that, Wei et al. (2022) optimized label-guided object localization by adjusting the trajectory of guiding lines.

For scene-level label annotation, Bell et al. (2001) projected 3D objects onto a 2D plane using a rectangular bounding box and merged the largest visible label rectangles, placing internal labels within and external labels without occlusion using a greedy algorithm. In AR scene labeling, Tenmoku et al. (2005) utilized a 3D model to enhance the visibility of objects in the real scene, using a textured 3D model for occluded objects and placing labels directly on target buildings. Makita et al. (2009) computed label positions based on object positions and shapes in wearable AR by minimizing an objective function to overlay labels accurately. Zhang et al. (2010) developed a method to register videos with 3D models and dynamically plan label placements in travel videos featuring prominent buildings or landmarks, contributing to scene-level labeling strategies. Hegde et al. (2020) used YOLOV2 to identify objects in the video stream and generate the corresponding labels, after which a saliency map was used to lay the labels in the non-significant space of the image. However, in AR glasses, the perception of these labels by users tends to decrease. Chen et al. (2023) presented a deep reinforcement learning-based approach designed to manage AR label placement in scenes involving moving objects. In the above approach, the user's viewpoint either remains stationary or rotates around the object, which limits the user's freedom of movement while navigating the virtual scene. Meanwhile, labels of some objects that are not in the field of view need to be displayed, in order to overcome this limitation and to maintain user depth perception, a force-based approach is used to optimize the label layout in the 3D scene.

2.3. Label filtering

Label filtering methods are designed to reduce the number of visible labels in the viewport, reducing occlusion and allowing the user to focus on useful information. Zhang and Sun (2005)'s system adjusted the label size according to the importance of the labels so as to optimize the user's visual experience and information acquisition, but its priority setting was performed manually. Peterson et al. (2009)'s label hierarchical approach improved readability and clarity by distributing labels at different depth levels, reducing the user's selection time when performing visual search tasks in visually overlap-intensive

environments. Tatzgern et al. (2013) applied combinatorial filtering and layout optimization techniques to create compact annotations by preparing compact visualizations from a set of discrete viewpoints, making them suitable for AR applications running on resource-limited mobile hardware. They used the textual information of the labels to cluster similar textual labels, and each cluster selected a representative label to be used for layout. Tatzgern et al. (2016) simplified labeling complexity in crowded scenarios by clustering similar items and selecting a representative representation of each cluster, creating a compact label layout. Clustering-based methods cause a label to be associated with multiple objects, which is more suitable for information retrieval but less desirable for localizing individual objects precisely. In addition, the clustering process relies on a large amount of a priori information, which makes its application relatively complex and inconvenient. Kouřil et al. (2018) presented a method for interactive labeling of crowded 3D scenes containing many instances of objects spanning multiple scales in size. They filtered labels for hierarchical scenes where many objects repeat, such as microbiological environments. The appropriate level of label was chosen by analyzing the scene's depth buffer and the scene objects' hierarchy tree. McNamara et al. (2019) used eye-tracking to determine user focus points, controlling label display by a fixed radius from an object to the annotation area. Simple filtering using the distance from the object to the center of view does not take into account the user's likely level of interest and is less effective in the localization process. Thus, our method uses the user's interest score of the object to filter the labels.

3. Method

When locating a target in a virtual scene, if there are multiple similar objects, it is hard to confirm whether the current observing object is the target, or multiple comparisons of similar possible objects may be required. Locating the target by observing the label can avoid this problem. Labels often carry specific semantic information, helping users quickly determine whether they are the label of the target object. Therefore, if the target label can be quickly found, the user can effectively improve the efficiency of locating the target object in the virtual scene. Given a set of objects with labels in a virtual scene and user viewpoint information, our method can optimize the label layout in real-time to reduce the user's time in finding the target label.

The pipeline of our method is shown in Fig. 2. The first step is to calculate the interest score of each object in the virtual scene. We compute the interest scores of the focused objects based on the duration that the user's line of sight is looking at the objects at a specific time. For other objects, we calculate the user's interest score by their similarity to the focused object, whose labels the user dwells on the most. We select only the labels of the n objects with the highest interest for display in the virtual scene. The second step is to estimate the user-perceived time of the above n labels using a mapping function of label attributes to user-perceived time. We obtain label attributes such as viewport position, visibility, and contrast, based on the label's location in the user's view and scene information. We use the random forest function fitted in the pilot user study to map the label attributes to the user's perceived time of the label. User-perceived time is the time interval from when the label is first displayed to when the user perceives the label. The third step is label layout optimization based on user perception force to generate the layout of these n visible labels. User perception force consists of two forces: fast-perception force and keep-in-view force. The direction of fast-perception force is determined by the gradient of the perceived time function, and the magnitude is determined by the object interest score. The magnitude of the keep-in-view force is proportional to the distance from the label to the viewport boundary in the viewport coordinate system, in the same direction as the camera's unit direction in world coordinates. This ensures that the labels are located within the user's viewport and placed in a position that is easier for the user to perceive. We have also retained the traditional force of the particle spring system.

In this section, we first introduce the concept of object interest scores (Section 3.1), and then explain how to select and fit a mapping function between the guided label attributes and their perceived time (Section 3.2). Finally, we describe the optimization of label layout based on the user perception force (Section 3.3) and a new metric (Section 3.4).

3.1. Object interest score

3.1.1. Definition

To quantify a user's real-time interest while navigating a scene, we introduce a definition called Object Interest Score $OIS(o,t)$, which measures the user's level of interest in object o in the virtual environment during a specific time interval t. $OIS(o,t)$ is an attribute of each object in the virtual environment. It initially has zero level of interest $OIS(o,0) = 0$ which means that initially, no object attracts the user's attention. As the user's viewpoint changes, $OIS(o,t)$ will be dynamically updated according to the change in the user's focus center, thus reflecting the change in the user's attention.

3.1.2. Computation

We classify the objects in the scene into two categories: the nearest object that intersects with the user's center line of sight under the current view and other objects. The nearest object that intersects with the center line of sight is the object that the user is currently observing, i.e., the object that the user is directly interested in at the time period t. The user's interest in the other objects depends on the degree of similarity between the object and the object that the user is interested in. Based on the above idea, we use different update strategies for the two types of objects.

For the current nearest object o_c in the user's center line of sight, the object interest score $OIS(o_c,t)$ is updated based on the proportion of time the object is being watched over the length of the entire time

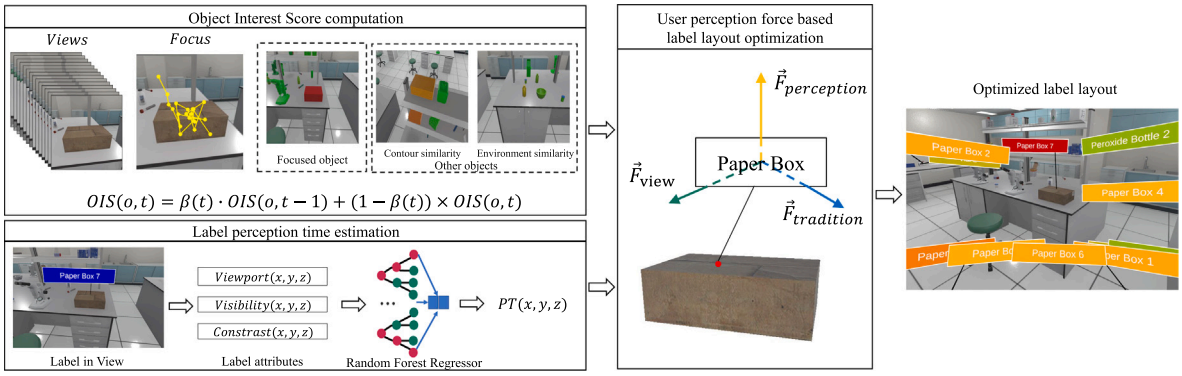


Fig. 2. The pipeline of user perception based label layout method.

interval. This approach directly reflects the intensity of the user's interest in the object based on the user's viewpoint. The interest score $OIS(o_c, t)$ can be calculated with Eq. (1).

$$OIS(o_c, t) = \frac{T(o_c, t)}{N} \tag{1}$$

where N is the total length of the time interval t, and $T(o_c, t)$ represents the cumulative time the object o_c intersects with the center line of user sight within the time interval.

For a given other object o_o, we compute $OIS(o_o, t)$ from o_c's interest score and similarity with o_c using Eq. (2).

$$OIS(o_o, t) = \frac{\Phi(o_o, o_c) \times OIS(o_c, t) + \sum_{i=1}^{n-1} \Phi(o_o, o_{c_i}) \times \frac{T(o_{c_i}, t)}{N}}{n} \tag{2}$$

where O_c is the nearest object in the user's central line of sight at the current point in time, and O_{c_i} is the nearest object other than O_c that intersects the user's central line of sight during time period t. n is the total number of objects that intersected the user's center line of sight during t. $\Phi(o_o, o_c(o_{c_i}))$ denotes the similarity between the two objects o_o and $o_c(o_{c_i})$. Our OIS is influenced by its visual similarity to other objects within the current field of view. As shown in Eq. (2), we compute the average similarity between object O_o and all other visible objects, rather than pairwise similarity with a specific object.

When assessing similarity, manual annotation methods often rely on attributes such as shape, color, and function to evaluate the resemblance between objects. However, this approach is both time-consuming and labor-intensive. To overcome these challenges, we employed two automated similarity calculation methods. In calculating the similarity between two objects $\Phi(o_1, o_2)$, we considered two aspects: contour similarity ϕ_c and environmental similarity ϕ_e (as shown in Eq. (3)). Square roots and squares are applied for numerical adjustments.

$$\Phi(o_1, o_2) = \frac{\phi_c(o_1, o_2)^2 + \sqrt{\phi_e(o_1, o_2)}}{2} \tag{3}$$

We evaluate contour similarity and environmental similarity by using the image pair comparison of two objects. In contour similarity estimation, we conduct a three-step comparison. The first step is image capture and preprocessing. Given an object, we place the camera directly above it, directly in front of it, and directly to the left of it at predetermined distances and capture its three images. The predetermined distance is set to ensure that the object is fully displayed on the image, and we used 0.6 m and 0.8 m for the two scenes in the user study. Once captured, these images are converted into grayscale images. Gaussian blurring is then performed to smooth the image. Adaptive thresholding is then performed to convert the grayscale image into a binary image. Finally, morphological operations (Serra, 1983) are applied to enhance the continuity of the major structures, ensuring that the images are optimally prepared for further similarity analysis. The second step is to extract the contours. For this purpose, we extracted object contours from each binary image (Suzuki et al., 1985). The third step is the comparison of the contours of the two objects. We compare the contours of two objects taken from corresponding viewpoints. To measure the similarity between these contours, we use Hu moments (Hu, 1962) to compute the similarity score, which is then normalized using the function $\phi = \frac{1}{1+\phi}$, scaling the values to the interval $(0, 1)$. The final contour similarity scores (ϕ_c) of the two objects are derived by averaging the normalized scores of contour similarity in three different view pairs.

In estimating the similarity of the environment of two objects, we removed the object from the scene, placed the viewpoint at the center of the object, and acquired images of the object in six directions. We then use the ORIENTED BRIEF (ORB) algorithm proposed by Rublee et al. (2011) to detect key points in the images and compute descriptors for the key points. The descriptors in the image pairs corresponding to the directions of the two objects are then matched using the Brute Force Hamming method. Cross-checking is performed during the matching process and only the matches that pass the ratio test are retained, thus filtering out matches with large distances. Finally, the similarity of the two images can be quantified by calculating the average distance d_{avg} and the maximum distance d_{max} of these matched images, and then the similarity is calculated using the function $\phi = 1 - d_{avg}/d_{max}$, scaling the values to the interval $(0, 1)$. The ϕ_e takes the maximum similarity calculated for the six directions.

3.1.3. State-transition-based model

After obtaining the user interest score of an object under the time period t, to further accurately capture and quantify the change in user interest over time, we adopt a state transition-based model (as shown in Eq. (4)) to update the user interest score of each object o in the scene again by adding the user interest scores of the objects from the time period $t-1$ to update $OIS(o, t)$ again for each object o in the scene.

$$OIS^*(o, t) = \beta(t) \cdot OIS^*(o, t-1) + (1 - \beta(t)) \cdot OIS(o, t) \tag{4}$$

where $OIS^*(o, t-1)$ represents the updated interest score from the previous time interval, and $\beta(t)$ is a decay factor that determines how interest levels transition over time. The decay factor $\beta(t)$ can be dynamically adjusted according to the user's behavioral patterns, reflecting the decay or enhancement of interest over time. If a user consistently focuses on the same object for several consecutive frames over a time period t, such that the object remains intersecting with the user's line of sight and is nearest, $\beta(t)$ decreases to sustain interest. Conversely, if the user's focus shifts often during t, reducing the reliability of interest, $\beta(t)$ increases to boost interest from the previous period. In addition, to prevent $OIS(o, t)$ from dropping to 0 too quickly when the user is in the gaze blank state, we set $\beta(t)$ to decay slowly over time. The decay factor $\beta(t)$ is computed in Eq. (5).

$$\beta(t) = \begin{cases} e^{-\lambda \times N} & \text{if } G_t = \emptyset \\ \beta_0 + (1 - \beta_0) \times \frac{SD(t)}{SD(t)+1} & \text{otherwise} \end{cases} \tag{5}$$

where λ denotes the decay constant, N is the total length of the time interval t, G_t denotes the set of focused objects within the time window t, β_0 is the base factor and $SD(t)$ is the dispersion of the user's central line-of-sight direction over the current time interval t. The $\frac{SD(t)}{SD(t)+1}$ is for normalisation, since $SD(t)$ is $(0, +\infty)$.

3.2. Label attributes - perceived time mapping

We propose a data-driven method for fitting a virtual scene-independent mapping function to model the relationship between label attributes and the user's perception time of a label in 3D space. This section first introduces the definition of mapping functions and then discusses the design of the pilot user study used for data collection, as well as the mapping function fitting methods using the collected data.

3.2.1. Mapping function definition

Schiefer et al. (2001) shows that human reaction time increases with the degree of field of view eccentricity, suggesting that labels further from the center of the visual field take more time to be noticed and recognized. Labels near the edge are more difficult to notice than those closer to the center. Thus, the relative position of a label in the field of view significantly impacts the user's reaction time. Additionally, the visibility of a label plays a key role in its perceivability and largely determines the user's reaction time.

In addition, Gervais et al. (1984) demonstrates that contrast significantly affects perception time. The contrast between a label and its background environment directly influences the user's perception speed and cognitive load. In 3D virtual environments, the color and brightness difference between labels and the background can notably impact reaction speed.

Based on these findings, we define the label attributes $A(l_o)$ of the label l_o of any object o as follows:

$$A(l_o) = (VP(x, y, z), C(x, y, z), V(x, y, z)).$$

The attributes of a label include the 3D location represented by $VP(x, y, z)$, label-environment contrast by $C(x, y, z)$, and label visibility by $V(x, y, z)$, where x, y, and z are the coordinates of the label l_o in the world coordinate system. These three attributes impact the user's perception time of the label in the 3D world.

Based on the label attributes, we define a function ($PT(x, y, z)$) that evaluates the user's perception time of the label in 3D space. The function is formulated as Eq. (6).

$$PT(x, y, z) = f(V(x, y, z), C(x, y, z), VP(x, y, z)) \tag{6}$$

where $PT(x, y, z)$ denotes the perception time for a label positioned at the 3D world coordinates (x, y, z). With the function $f(A(l_o))$, we can predict the probable perception time of the user. Unlike the typical empirical formula, we use data fitting to determine it. The next section describes how to obtain f.

3.2.2. Pilot user study design: mapping function selecting and fitting

The pilot user study focuses on fitting and selecting a mapping function from label attributes to the user-perceived time of labels. We used random forest (RF), support vector machine (SVM), and decision tree (DT) regressors to fit the collected data by recording the user-perceived time of label attributes and corresponding labels. Finally, we chose the final mapping function based on the minimum root mean square error ($RMSE$). The details of the experiment are as follows.

Participants. We have recruited 10 participants, 5 males, and 5 females, between 20 and 31 years old. 6 of our participants had prior VR experience. Participants had normal or corrected vision, and none reported vision or balance disorders.

Hardware and Software Setup. The experiment utilized HTC Cosmos VR headsets with Droolon F1 eye-tracker, and the Unity engine was used for all virtual scene development and experimental procedures. The setup included a computer with a 3.79GHz Intel(R) Core(TM) i7-10700KF CPU, 24GB RAM, and an NVIDIA GeForce GTX 3080 Ti graphics card. The tracking area for the headset was set to a physical space of 3 m × 3 m.

Task. In the virtual environment, the system generates randomly colored labels every 1–10 s at different locations in 3D space. Participants were asked to gaze at as many of these labels as possible from a predetermined location, i.e., the gaze point landed on these labels, and the labels that the user gazed at disappeared immediately. If a label had never been gazed at by the user, it was removed by the system after 10 s.

Data collection. The system records the attributes of disappearing labels and the time interval between their appearance and disappearance. The methods for obtaining or calculating perception time $PT(x, y, z)$, visibility $V(x, y, z)$, contrast $C(x, y, z)$, and viewport coordinates $VP(x, y, z)$ are as follows:

- $PT(x, y, z)$ records the time interval from the time the label was generated to the time it was first viewed by the participant, which is recorded as the perceived time of the label. The value of PT for unseen labels was set to 10s.
- $V(x, y, z)$ was measured by sampling the outline of the label, projecting rays from the camera angle to the sampling points, and assessing the proportion of accessible sampling points to the total sampling points.
- $C(x, y, z)$ is calculated based on the contrast between the label and its background in a 50x50 pixel area. To account for multiple visual factors in a unified contrast score, the system converts the label color and the background color from RGB to HSV format and calculates the absolute difference between the two colors in terms of hue, saturation, and brightness. Finally, a weighted contrast score is assigned based on these differences to quantify the contrast between the colors.
- $VP(x, y, z)$ is converted from world coordinates (x, y, z) to viewport coordinates.

Mapping Function Fitting and Selecting. We collected a set of valid data and, following a common data allocation strategy, used 70% of the data as a training set to fit the RF, SVM, and DT regressors. The remaining 30% was used as a validation set to compute $RMSE$ of the three fitted models. $RMSE$ was 3.55 for DT regressor, 2.79 for RF regressor, and 3.87 for SVM regressor. These results indicate that the random forest regressor RF performs the best in this pilot user study, and was therefore selected as the mapping function from label attributes to user-perceived time of labels.

We visualized the fitted RF, SVM and DT regressors. Fig. 3 shows the relationship between the 3D viewport position and the label perceived time with the contrast and visibility of the labels both being 0.5. The three axes represent different 3D coordinates, and the color represents the user perception time of the label; the bluer the color, the shorter the perception time, and vice versa. From the visualization of the fitted RF regressor, it can be seen that the labels placed in the center region of the viewport all have a shorter perception time than the labels placed in the peripheral region of the same depth. The perception time of the labels placed close to the user's viewpoint is less than that of the labels placed far away. Both of these laws are consistent with the biological laws of visual perception (Schiefer et al., 2001), as well as with our intuition.

From the visualization of the fitted SVM regressor, only the law that the label perceived time becomes larger as the distance from the user's viewpoint increases can be seen, while the perception time of labels in the central region is similar to that of labels in the peripheral region, which is not in line with the biological law of visual perception (Schiefer et al., 2001), as well as our intuitive sense.

In contrast, the results from the DT regression model are less obviously regular. The visualizations of the fitted DT models did not show strong or consistent patterns, making it difficult to identify any significant relationship between label attributes and perceived time. This lack of clarity may suggest that decision tree models are less effective

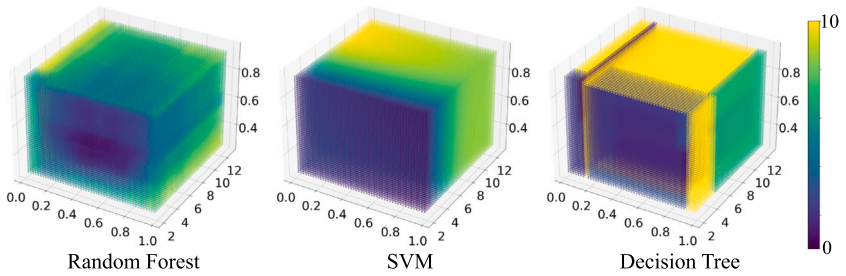


Fig. 3. Visualization results of the decision tree regressor, SVM regressor, and random forest regressor.(contrast = 0.5, visibility = 0.5). y-axis from bottom to top, x-axis from left to right, z-axis from front to back. (0.5,0.5,0) is the center of the field of view.

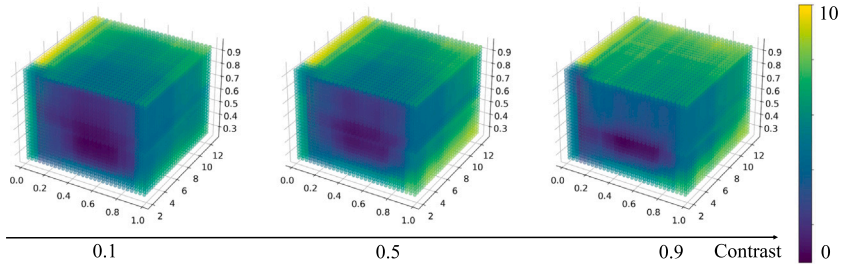


Fig. 4. Visualization results of the random forest regressor.(contrast =0.1,0.5,0.9 visibility = 0.5). y-axis from bottom to top, x-axis from left to right, z-axis from front to back. (0.5,0.5,0) is the center of the field of view.

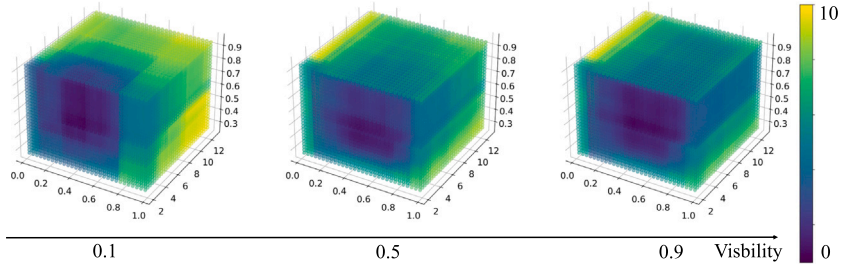


Fig. 5. Visualization results of the random forest regressor.(contrast =0.5, visibility = 0.1,0.5,0.9). y-axis from bottom to top, x-axis from left to right, z-axis from front to back. (0.5,0.5,0) is the center of the field of view.

at capturing the complex dynamics of how label position and distance affect user perceived time.

Fig. 4 shows the visualization of the Random Forest regressor with a visibility of 0.5 and contrasts of 0.1, 0.5 and 0.9. By comparing the range of colors in the central region on the front side of the viewport, we can clearly observe that the blue region on the front side shrinks accordingly as the contrast increases. This phenomenon suggests that although one intuitively believes that 'the higher the contrast, the easier the label is perceived', the increase in contrast actually leads to a longer label perception time. This finding contradicts our initial expectation, but is consistent with the research literature of Stoyanova et al. (1987); Bruno and Johnston (2010); Benton and Redfern (2016). Notably Benton and Redfern (2016) showed that perception time increases with contrast, but only by a small amount.

Fig. 5 shows the visualization results of the Random Forest regressor with a contrast of 0.5 and visibility settings of 0.1, 0.5 and 0.9. Comparing the image with a visibility of 0.1 to the images with visibilities of 0.5 and 0.9, a clear trend can be observed: the blue area at the front of the viewport widens and deepens as the visibility increases. In addition, when comparing labels at the same depth in the viewport, it can be observed that the blue color at the same depth becomes more obvious as the visibility increases. This phenomenon suggests that beyond a certain depth, the perception time of a label increases significantly with visibility. This finding is consistent with common sense.

3.3. User perception based label layout

Marks and Shieber (1991) points out that the label layout problem is often NP-hard, and in order to dynamically optimize the layout of the guided labels in real time, we use the force-based particle-spring model, a technique inspired by Hartmann et al. (2004), to solve the optimization problem by physical simulation approximation. In this model, labels are treated as particles constrained by springs connected by fixed anchors, and label positions are optimized by simulating the interaction forces between labels.

In our particle spring system, the forces acting on the labels are divided into two main categories: one is our newly proposed user perception force, which enables the labels to be placed at 3D spatial locations that the user can quickly perceive. User perception force consists of two forces: fast-perception force $\vec{\mathbf{F}}_{\text{perception}}$ and keep-in-view force $\vec{\mathbf{F}}_{\text{view}}$. The other is the traditional dynamic adjustment forces used in existing methods, such as Hartmann's 2D technique Hartmann et al. (2004). These adjustment forces effectively prevent common annotation issues such as annotation occlusion, annotation deviation from the target object, and crossing of guide lines. However, since Hartmann's method operates in 2D space, while our goal is to optimize annotation layout in 3D, we extend these dynamic adjustment forces from 2D to 3D. A damping force inversely proportional to the direction of the velocity is also introduced to maintain temporal coherence.

3.3.1. Fast-perception force

Fast-perception force $\vec{\mathbf{F}}_{\text{perception}}$ is introduced to guide the movement of labels to 3D spatial locations where the user's perceived time is minimized, by utilizing the gradient of the mapping of label attributes to perceived time. The direction of $\vec{\mathbf{F}}_{\text{perception}}$ is determined by the gradient of the user's perceived time function, and the magnitude is determined by the object interest score.

As described earlier in Section 3.2, for a given label and its attributes, the user perception time function $PT(x, y, z)$ for the label is modeled using a random forest regressor. Since the function is implicit, it is more difficult to compute the gradient directly. Therefore, we approximate the gradient by calculating the change in user label perceived time in fixed increments along the coordinate axes. We present the x axis as an example, and y, z axis gradients are computed in the same way as the x-axis.

$$PT_{x+\Delta x} = f(V(x+\Delta x, y, z), C(x+\Delta x, y, z), VP(x+\Delta x, y, z)) \tag{7}$$

$$PT_{x-\Delta x} = f(V(x-\Delta x, y, z), C(x-\Delta x, y, z), VP(x-\Delta x, y, z)) \tag{8}$$

$$PT_{x} = f(V(x, y, z), C(x, y, z), VP(x, y, z)) \tag{9}$$

Then, the positive gradient, PT_{x+}, and the negative gradient, PT_{x-}, are computed as follows:

$$PT_{x+} = \frac{PT_{x+\Delta x} - PT_{x}}{\Delta x} \tag{10}$$

$$PT_{x-} = \frac{PT_{x-\Delta x} - PT_{x}}{\Delta x} \tag{11}$$

The x-axis component D_x of the direction vector of the perception force $\vec{\mathbf{F}}_{\text{perception}}$ can then be determined:

$$D_x = \begin{cases} -PT_{x+} & \text{if } PT_{x+} < 0 \text{ and } (PT_{x-} \geq 0 \text{ or } |PT_{x+}| > |PT_{x-}|) \\ PT_{x-} & \text{if } PT_{x-} < 0 \text{ and } (PT_{x+} \geq 0 \text{ or } |PT_{x-}| \geq |PT_{x+}|) \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

This strategy prioritizes the gradient descent direction (with a larger negative slope amplitude) and ignores ambiguous gradients or positive gradients, thereby ensuring a more stable force direction during the optimization process. The reference direction vector of the perception force is $\vec{D} = (D_x, D_y, D_z)$.

To maintain temporal coherence in label placement, which is essential for minimizing abrupt changes in label positions from one frame to the next, we employ a method of temporal smoothing. Since $\vec{\mathbf{F}}_{\text{perception}}$ needs to be computed in each frame, there may be abrupt changes in the direction of $\vec{\mathbf{F}}_{\text{perception}}$ between frames, making the label placement unstable, so we perform a temporal smoothing of the direction. We compute the direction vector of the current frame $\vec{D}_t$ by spherical linear interpolation using the direction vector $\vec{D}_{t-1}$ of the previous frame and $\vec{D}$, as in Eq. (13).

$$\vec{D}_t = \frac{\sin((1-\alpha)\theta)}{\sin\theta}\vec{D}_{t-1} + \frac{\sin(\alpha\theta)}{\sin\theta}\frac{\vec{D}}{\|\vec{D}\|} \tag{13}$$

$$\theta = \cos^{-1}\left(\vec{D}_{t-1} \cdot \frac{\vec{D}}{\|\vec{D}\|}\right) \tag{14}$$

$$\alpha = 1 - 0.9 \cdot \frac{\theta}{\pi} \tag{15}$$

Where θ is the angle between $\vec{D}_{t-1}$ and $\vec{D}$, and α is an interpolation coefficient controlled by θ.

Thus, the perception force applied to each label at time t is formulated as Eq. (16):

$$\vec{\mathbf{F}}_{\text{perception}}(t) = k_{\text{perception}} \times OIS(t) \times \vec{D}_t \tag{16}$$

Where $k_{\text{perception}}$ is a user-settable factor.

3.3.2. Keep-in-view force

When using the fitted user-perceived time mapping to predict the possible perception time of the label, the extracted label attributes need to be located within the user's field of view. Therefore $\vec{\mathbf{F}}_{\text{view}}$ is introduced to ensure that the label stays within the user's field of view, preventing the label from moving outside the viewport. The magnitude of $\vec{\mathbf{F}}_{\text{view}}$ is proportional to the distance from the label to the viewport boundary in the viewport coordinate system, in the same direction as the camera's unit direction in world coordinates. $\vec{\mathbf{F}}_{\text{view}}$ can be calculated with Eq. (17).

$$\vec{\mathbf{F}}_{\text{view}} = k_{view}\left(\vec{\mathbf{F}}_x + \vec{\mathbf{F}}_y + \vec{\mathbf{F}}_z\right) \tag{17}$$

Where k_{view} is a user-settable factor, and the components $\vec{\mathbf{F}}_x$, $\vec{\mathbf{F}}_y$, and $\vec{\mathbf{F}}_z$ of $\vec{\mathbf{F}}_{\text{view}}$ in the x, y, and z directions are computed with Eqs. (18)–(20), respectively.

$$\vec{\mathbf{F}}_x = \begin{cases} (m_x - x) \cdot \vec{\mathbf{i}} & \text{if } x < m_x \\ (x - (1 - m_x)) \cdot (-\vec{\mathbf{i}}) & \text{if } x > 1 - m_x \\ 0 & \text{otherwise} \end{cases} \tag{18}$$

$$\vec{\mathbf{F}}_y = \begin{cases} (m_y - y) \cdot \vec{\mathbf{j}} & \text{if } y < m_y \\ (y - (1 - m_y)) \cdot (-\vec{\mathbf{j}}) & \text{if } y > 1 - m_y \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

$$\vec{\mathbf{F}}z = \begin{cases} (m_{z_{\min}} - z) \cdot \vec{\mathbf{k}} & \text{if } z < m_{z_{\min}} \\ (z - m_{z_{\max}}) \cdot (-\vec{\mathbf{k}}) & \text{if } z > m_{z_{\max}} \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

$\vec{\mathbf{i}}$, $\vec{\mathbf{j}}$, and $\vec{\mathbf{k}}$ are unit vectors in the right, up, and forward directions, respectively, of the camera coordinate system. $\vec{\mathbf{i}}$ represents the unit vector along the x-axis, pointing to the right of the camera. $\vec{\mathbf{j}}$ represents the unit vector along the y-axis, pointing upward from the camera, and is used to adjust the vertical position of the label. $\vec{\mathbf{k}}$ is the unit vector along the z-axis, pointing towards the front of the camera, and controls the label's position in the depth direction. The margins m_x, m_y, $m_{z_{\min}}$, and $m_{z_{\max}}$ define the critical distances in the x, y, and z directions, respectively, within which the force becomes active, determining the amount of adjustment required when the label moves away from the viewport.

3.3.3. Label position updating

For each label, the force $\vec{\mathbf{F}}$ is the sum of the user perception force and the traditional dynamic adjustment forces, which include six forces: label spring force $\vec{\mathbf{F}}_{\text{spring}}$, area correction force $\vec{\mathbf{F}}_{\text{area}}$, label repulsion force $\vec{\mathbf{F}}_{\text{label}}$, obstacle repulsion force $\vec{\mathbf{F}}_{\text{obstacle}}$, damping force $\vec{\mathbf{F}}_{\text{damping}}$ and leading-line cross force $\vec{\mathbf{F}}_{\text{cross}}$. Therefore, $\vec{\mathbf{F}}$ can be computed with Eq. (21).

$$\vec{\mathbf{F}} = \vec{\mathbf{F}}_{\text{perception}} + \vec{\mathbf{F}}_{\text{view}} + \vec{\mathbf{F}}_{\text{label}} + \vec{\mathbf{F}}_{\text{spring}} + \vec{\mathbf{F}}_{\text{area}} + \vec{\mathbf{F}}_{\text{damping}} + \vec{\mathbf{F}}_{\text{obstacle}} + \vec{\mathbf{F}}_{\text{cross}} \tag{21}$$

After calculating the force on each label, we calculate the acceleration and velocity according to Newton's second law to finally obtain the 3D spatial location of the label placement.

3.4. Label perceived time cost

In order to quantitatively evaluate the effectiveness of our approach in this paper, we also propose an objective metric of label perceived time cost $Cost_p$ that focuses on evaluating whether the object labels that are of high interest to the user are placed in the position with shorter perception time among the layouts of all the visible labels at the moment t. Specifically, the cost metric reaches a minimum when the interest score $OIS(o,t)$ of an object reaches a peak value and its corresponding label is located at the position with the shortest perceived time $PT(x,y,z)$. Therefore, the perceived time cost metric $Cost_p$ can be calculated by Eq. (22).

$$Cost_p(t) = \sum_{i=1}^{n}(1 - OIS(o_i,t)) \times PT(x_i,y_i,z_i) \tag{22}$$

where n denotes the number of visible labels at moment t.

4. User study

To evaluate the effectiveness of our proposed label layout method, we conducted a user study. We compare our method with the traditional dynamic force-based layout method (CC_1) proposed by Hartmann et al. (2004) and use the discrete update strategy used by Tatzgern et al. (2016) when the user's moving distance or rotation angle exceeds a threshold to ensure temporal coherence (CC_2), as well as the filtering method (CC_3) based on the distance from the center of the view proposed by McNamara et al. (2019). In CC_1, CC_2, and CC_3, labels that are occluded by an object are displayed with the occluded part dimmed. As shown in Fig. 6, it addresses the common problem of labels becoming unreadable when they pass behind 3D objects in the scene. The user can still read the entire label but intuitively understands which part is currently hidden. In addition, our approach covers the case where fast-perception force is not used (EC_1) and the case where fast-perception force is used (EC_2). In the CC_3 setup, we show only the 15 labels closest to the center of view. In the EC_1 and EC_2 setups, we show only the top 15 object labels with the highest interest. In all conditions, the visible labels are oriented towards the user, as shown in Fig. 6. In this user study design, we note that EC_1 and EC_2 methods use color as an additional visual channel to enhance the discernibility of OIS labels, while CC does not. This suggests that the difference in task completion

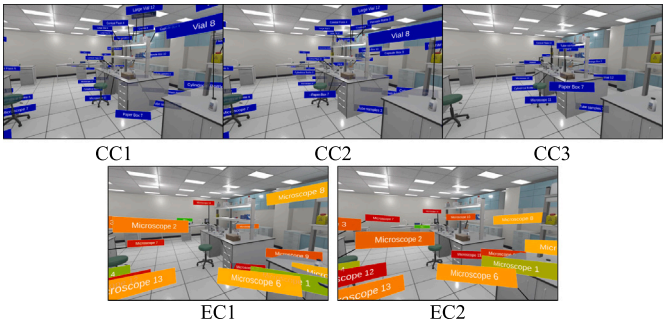


Fig. 6. Comparison of 3D label layout Methods. **(a)** CC_1: Traditional dynamic force-based method Hartmann et al. (2004). **(b)** CC_2: CC_1+Discrete update strategy ensuring temporal coherence Tatzgern et al. (2016). **(c)** CC_3: CC_1+Distance-based filtering method McNamara et al. (2019). **(d)** EC_1: Our approach without perception force. **(e)** EC_2: Our approach with perception force.

time between EC_2 and CC methods may stem from the combined effects of both the layout algorithm and color (which is the visualization of the OIS factor). However, EC_1 and EC_2 use the same visual style, with the layout algorithm being the only variable, so the performance difference can be clearly attributed to the advantages of our proposed fast-perception layout. The research was performed under the oversight of the Biology and Medical Ethics Committee of Beihang University, with protocol number BM20240277. Consent from the human subjects in the research was obtained.

4.1. Participants

This study recruited 16 participants, including 8 males and 8 females, aged between 20 and 30, through social media platforms. Ten of these participants had experience using VR equipment. All participants had normal or corrected-to-normal vision and no impairments in vision or balance. All participants took part in five conditions.

All participants were adults and provided written informed consent before the experiment began. To prevent the risk of motion sickness interfering with the real environment, the experiment was conducted in a well-equipped VR laboratory with an open, accessible area. Participants had ample time to familiarize themselves with the VR equipment before the experiment began and could terminate the experiment at any time. No personally identifiable information was collected in this study, and all experimental data were anonymized and used solely for scientific research purposes.

4.2. Task and procedure

The task provides a local snapshot of the object that the user needs to locate and its environment, and gives the name of the object. The participant is asked to find a specific labeled object in the scene as quickly as possible. We created two different scenes: an indoor chemistry lab scene (S_1) and an outdoor courtyard scene (S_2). As shown in Fig. 7, S_1 is shown on the left and S_2 on the right. In each scene, the objects are randomly distributed, and identical items are assigned unique labels using sequence numbers. In S_1, we placed 127 items of 15 different types on tables and shelves. The user navigates the scene through a combination of joystick control and body movement to complete the localization task. In S_2, we set up three houses as obstacles and randomly placed 105 items of 16 different types in the courtyard. In addition to body movement, users used a “teleportation” technique by pressing the trigger button on the controller to explore the area.

Before the experiment started, participants underwent a three-minute comprehensive training session in a simple scene to familiarize themselves with the five different label layout methods.



Fig. 7. Left: Indoor chemistry laboratory (S_1). Right: Outdoor courtyard (S_2). Objects searched for are marked in blue.

In S_1 and S_2, participants are required to search for 12 and 16 given target objects, respectively. The order of the five conditions was randomized across participants using a Latin square design. Throughout the experiments involving different methods, the sequence in which target objects appeared was randomized. Each participant completed 140 searches (5conditions × 12searches in S_1 + 5conditions × 16searches in S_2), with a 30s rest between each search. For each search, participants first see a snapshot of the current target object and its corresponding label. Once the participant confirms the target object, the system switches to the virtual scene. The participant then observes the objects and labels, finds the target in the current codification, points a raycast to the target object, and presses a confirmation button to complete the search. After completing all searches (12 times for S_1 and 16 times for S_2) under each label layout method condition (e.g., one task), participants filled out questionnaires (including NASA-TLX, perceived time cost, and usability evaluation) immediately. For the SSQ questionnaire, they were asked to complete the pre-score before the task began.

4.3. Metrics

We used three objective metrics: task completion time (in seconds), user movement distance (in meters), and user rotation angle (in degrees). Each search task recorded the time from when the user entered the search scene to when the user found the target and pressed the confirmation button, the sum of the distance traveled in the process, and the sum of the head rotation angle. Finally, the metric values of all search tasks in a single scene were summed for these metrics. In addition, we used our newly proposed label perceived time cost for the five conditions as an objective metric. In the first search task of S_1, we recorded the degree of interest of visible labels during the process of a participant’s gaze from not gazing at an object to gazing at an object (about 700 frames), and the perception time cost was calculated using Eq. (22).

We used three subjective metrics to evaluate users’ VR experience: (1)the NASA TLX questionnaire introduced by (Hart, 2006), assessing workload across five aspects (mental, temporal, performance, effort, and frustration) with a 0–10 scoring scale; (2)a usability questionnaire to evaluate users’ perceptions of technology usability, focusing on intuition, efficiency, accuracy, ease-of-use, naturalness, and satisfaction, also scored from 0–10; (3)the Simulator Sickness Questionnaire introduced by (Kennedy et al., 1993), identifying potential discomfort with a 0–10 scoring scale.

4.4. Hypotheses

To compare the performance of our method with the traditional method, we propose the following hypotheses:

H1: Compared to CC_1, CC_2, CC_3, and EC_1, the user can locate the target more efficiently with EC_2.

H2: Compared to CC_1, CC_2, CC_3, and EC_1, the viewpoint translation distance is reduced using EC_2.

H3: Compared to CC_1, CC_2, CC_3, and EC_1, the viewpoint rotation angle is reduced using EC_2.

H4: Compared to CC_1, CC_2, CC_3, and EC_1, EC_2 has a lower perception time cost.

H5: Compared to CC_1, CC_2, CC_3, and EC_1, users experience lower fatigue when locating target objects using EC_2.

H6: Compared to CC_1, CC_2, CC_3, and EC_1, EC_2 is easier to use.

H7: Compared to CC_1, CC_2, CC_3, and EC_1, EC_2 has lower post-SSQ score.

4.5. Statistical analysis

For each metric, we used a one-way repeated measures ANOVA to compare the results under four control conditions (CC_1, CC_2, CC_3, EC_1) and one experimental condition (EC_2). First, the Shapiro-Wilk test was used to verify the assumption of normal distribution of the data. Then, the Mauchly's test assessed the assumption of sphericity. If the assumption of sphericity was violated, the Greenhouse-Geisser correction was applied. Following the ANOVA, if the null hypothesis (i.e., no significant difference between conditions) was rejected ($p < 0.05$), Bonferroni correction was applied for post-hoc comparisons. For time-dependent variables, we also quantified the effect size using Cohen's d (Cohen, 2013). Statistical analyses were performed using SPSS software (IBM, n.d.). d values were translated into qualitative estimates of effect size, including *huge* ($d > 2.0$), *very large* ($2.0 > d > 1.2$), *large* ($1.2 > d > 0.8$), *medium* ($0.8 > d > 0.5$), *small* ($0.5 > d > 0.2$) and *very small* ($0.2 > d > 0.01$).

4.6. Results

Table 1 provides the task completion time for five conditions in two scenes. Statistical significance is indicated by asterisks. For S_1, the assumption of sphericity was verified: $p = 0.243$. The overall ANOVA revealed significant differences between the five conditions: ($F_{4,28} = 14.733, p < 0.001$). For S_2, the assumption of sphericity was verified: $p = 0.164$. The overall ANOVA revealed significant differences between the five conditions: ($F_{4,28} = 21.087, p < 0.001$). In S1, post-hoc analysis shows that EC_2 significantly reduces task completion time compared to CC_1 ($p < 0.001$), CC_2 ($p = 0.001$), and CC_3 ($p = 0.001$), with effect sizes ranging from $VeryLarge$ to $Huge$. Compared to EC_1, the reduction in task completion time is also significant ($p = 0.017$) with a $Large$ effect size. In S2, post-hoc analysis shows that EC_2 significantly reduces task completion time compared to CC_1 ($p < 0.001$), CC_2 ($p < 0.001$), and CC_3 ($p < 0.001$), with effect size $Huge$. Compared to EC_1, the reduction in task completion time is also significant ($p = 0.025$) with a $Large$ effect size. This result supports ***H1***.

Table 2 presents the viewpoint translation distances moved by participants for five conditions in two scenes. Statistical significance is denoted by asterisks. For S_1, the assumption of sphericity was violated with $p < 0.001$. After applying the Greenhouse-Geisser correction, the overall ANOVA revealed significant differences between the five conditions ($F_{1.204,8.430} = 15.409$, $p = 0.03$). For S_2, the assumption of sphericity was violated ($p = 0.004$). After applying the Greenhouse-Geisser correction, the overall ANOVA still indicated significant differences between the five conditions ($F_{1.851,12.956} = 29.100$, $p < 0.001$). Post-hoc analysis indicates that EC_2 significantly reduces the viewpoint translation distance compared to CC_1 ($p < 0.001$), CC_2 ($p < 0.001$), and CC_3 ($p < 0.001$) in both scenes, with a $Huge$ effect size. Compared to EC_1, EC_2 reduces the viewpoint translation distance, but the difference is not significant ($p = 0.092$ for S_1, $p = 0.213$ for S_2. The effect sizes of the reduction are $Large$ (S_1) and $Medium$ (S_2). This result supports ***H2***.

Table 3 gives the viewpoint rotation angles for five conditions in two scenes. Statistical significance is indicated by an asterisk. For S_1, the sphericity assumption was violated: $p = 0.007$. After applying the Greenhouse-Geisser correction, the overall ANOVA revealed a significant difference between the five conditions: ($F_{1.687,11.807} = 25.359, p < 0.001$). For S_2, the sphericity assumption was verified: $p = 0.119$. The overall ANOVA revealed a significant difference between the five conditions: ($F_{4,28} = 33.310, p < 0.001$). Post-hoc analysis indicates that EC_2 significantly reduces the viewpoint rotation angles compared to CC_1 ($p < 0.001$ for S_1, $p < 0.001$ for S_2), CC_2 ($p = 0.001$ for S_1, $p < 0.001$ for S_2), and CC_3 ($p = 0.014$ for S_1, $p < 0.001$ for S_2) across both scenes, with effect sizes ranging from $VeryLarge$ to $Huge$. Compared to EC_1, EC_2 reduces the viewpoint rotation angle, but the difference is not significant ($p = 0.098$ for S_1, $p = 0.201$ for S_2. The effect sizes of the reduction are $Large$ (S_1) and $Medium$ (S_2). This result supports ***H3***.

Fig. 8 shows the label perceived time cost for five conditions. Starting from the focus of the line of sight to the target, CC_2 remains unchanged after stabilization during the process, while CC_3 changes very little after stabilization, while the cost curves of CC_1, EC_1, and EC_2 gradually converge to the stable value from the start frame to the end frame. The perception time costs of CC_1 and CC_2 are significantly ($p < 0.001$) higher than those of EC_2. The value of CC_3 and EC_1 is lower than that of EC_2 after stabilization, and the difference is significant (after 200 frames, $p < 0.001$). This result supports ***H4***.

The NASA-TLX scores are presented in Fig. 9 of five conditions. We examined the assumption of sphericity for the five NASA-TLX dimensions, with the following results: mental ($p = 0.036$), temporal

Table 1
Task completion time to locate targets, in seconds.

Task	Condition	Avg ± std. dev.	$\lvert EC_2\text{-}XC_i\rvert / XC_i$	p	Cohen's d	Effect size
S_1	CC1	969.65 ± 221.61	72.3%	< 0.001*	4.16	Huge
	CC2	483.63 ± 149.78	44.5%	0.001*	1.75	Very Large
	CC3	525.17 ± 179.35	48.9%	0.001*	1.82	Very Large
	EC1	396.77 ± 149.62	32.4%	0.017*	1.05	Large
	EC2	268.26 ± 88.21	–	–	–	–
S_2	CC1	712.75 ± 127.98	55.3%	< 0.001*	3.53	Huge
	CC2	619.16 ± 158.30	48.6%	< 0.001*	2.32	Huge
	CC3	734.45 ± 142.66	56.6%	< 0.001*	3.46	Huge
	EC1	423.45 ± 145.65	24.8%	0.025*	0.86	Large
	EC2	318.52 ± 92.45	–	–	–	–

Table 2
The viewpoint translation distances to locate targets, in meters.

Task	Condition	Avg ± std. dev.	$\lvert EC_2\text{-}XC_i\rvert / XC_i$	p	Cohen's d	Effect size
S_1	CC1	321.31 ± 139.41	84.8%	< 0.001*	2.76	Huge
	CC2	227.72 ± 53.31	78.5%	< 0.001*	4.70	Huge
	CC3	289.85 ± 105.27	83.1%	< 0.001*	3.23	Huge
	EC1	55.11 ± 7.68	11.3%	0.092	0.84	Large
	EC2	48.86 ± 7.22	–	–	–	–
S_2	CC1	1727.37 ± 589.09	72.5%	< 0.001*	2.94	Huge
	CC2	1784.39 ± 179.17	73.4%	< 0.001*	8.39	Huge
	CC3	1816.85 ± 321.56	73.9%	< 0.001*	5.48	Huge
	EC1	578.67 ± 186.63	18.0%	0.213	0.65	Medium
	EC2	474.72 ± 128.95	–	–	–	–

Table 3
The viewpoint rotation angles to locate targets, in degrees.

Task	Condition	Avg ± std. dev.	Mean reduction	p	Cohen's d	Effect size
S_1	CC1	31164.64 ± 9989.52	83.5%	< 0.001*	3.64	Huge
	CC2	9714.49 ± 3051.51	46.9%	0.001*	1.88	Very Large
	CC3	11181.53 ± 6243.29	53.9%	0.014*	1.32	Very Large
	EC1	7071.46 ± 2897.60	27.1%	0.098	0.82	Large
	EC2	5156.19 ± 1553.92	–	–	–	–
S_2	CC1	23237.22 ± 5929.00	78.3%	< 0.001*	4.20	Huge
	CC2	24458.44 ± 4637.15	79.4%	< 0.001*	5.61	Huge
	CC3	23153.33 ± 3422.69	78.2%	< 0.001*	6.75	Huge
	EC1	6942.04 ± 3687.99	27.5%	0.201	0.67	Medium
	EC2	5036.37 ± 1641.81	–	–	–	–

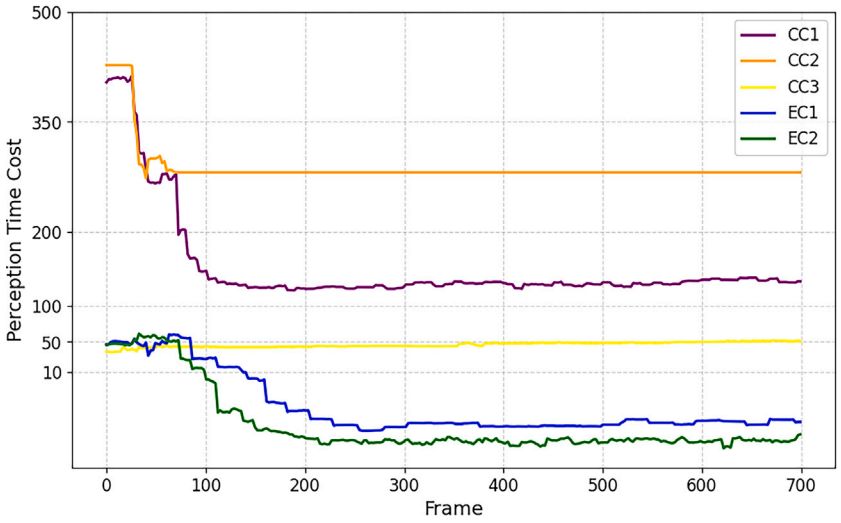


Fig. 8. Perception time cost over frames.

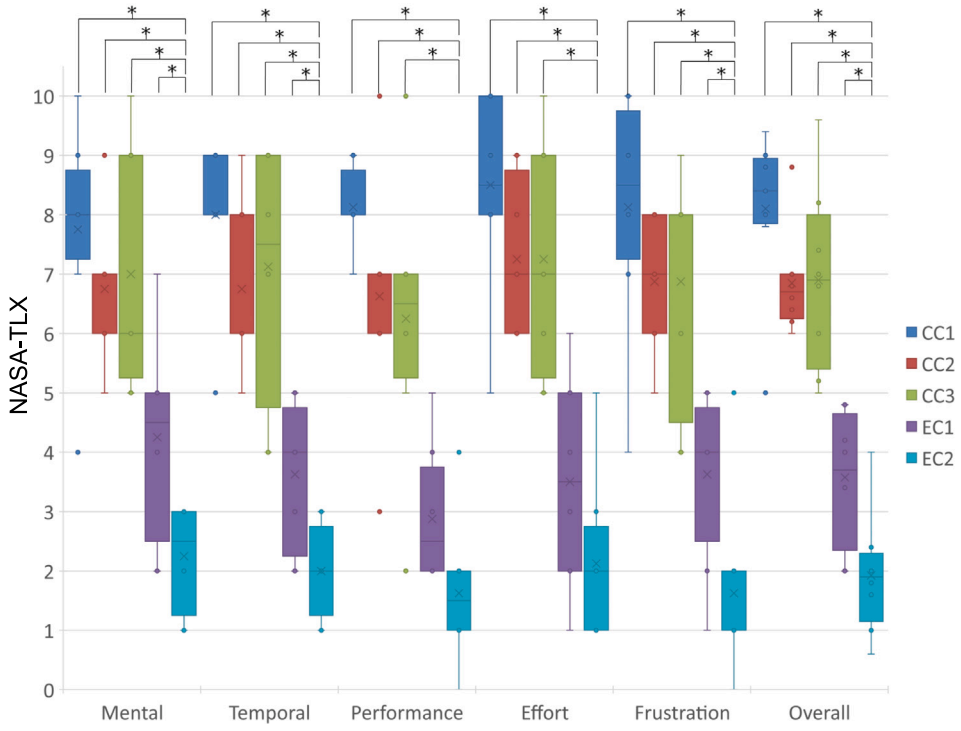


Fig. 9. NASA-TLX scores for five questions and overall. Significant differences are denoted with an asterisk.

($p = 0.230$), performance ($p = 0.049$), effort ($p = 0.442$) and frustration ($p = 0.099$), as well as the overall score ($p = 0.139$). The p-values for mental and performance were below 0.05, thus violating the assumption of sphericity, and the Greenhouse-Geisser correction was applied. The overall results of the ANOVA showed statistically significant differences between conditions. The results were as follows: mental ($F_{1.717,12.016} = 17.458, p < 0.001$), temporal ($F_{4,28} = 23.739, p < 0.001$), performance ($F_{1.513,10.589} = 24.780, p < 0.001$), effort ($F_{4,28} = 22.328, p < 0.001$), frustration ($F_{4,28} = 20.071, p < 0.001$), and overall score ($F_{4,28} = 31.290, p < 0.001$), showed statistically significant differences in all measures. Post hoc analyses showed that EC_2 was significantly lower than CC_1 ($p < 0.001$), CC_2 ($p < 0.001$), and CC_3 ($p < 0.001$) on all five dimensions and the overall score. Compared to EC_1, EC_2 was significantly lower on mental ($p = 0.01$), temporal ($p = 0.006$), frustration ($p = 0.016$), and overall scores ($p = 0.008$). This result supports ***H5***.

The usability scores of five conditions were compared in Fig. 10. We investigated the sphericity assumption across six aspects of usability and found that all aspects met this assumption: accuracy ($p = 0.164$), intuition ($p = 0.104$), efficiency ($p = 0.251$), naturalness ($p = 0.09$), satisfaction ($p = 0.143$), and ease-of-use ($p = 0.097$), with an overall score of ($p = 0.201$). The adjusted ANOVA revealed significant differences among five conditions in all aspects: intuition ($F_{4,28} = 30.938, p < 0.001$), efficiency ($F_{4,28} = 32.407, p < 0.001$), accuracy ($F_{4,28} = 19.353, p < 0.001$), naturalness ($F_{4,28} = 9.484, p < 0.001$), satisfaction ($F_{4,28} = 32.611, p < 0.001$), ease-of-use ($F_{4,28} = 28.930, p < 0.001$), and the overall score ($F_{4,28} = 34.670, p < 0.001$). For the subscores, the post-hoc analyses

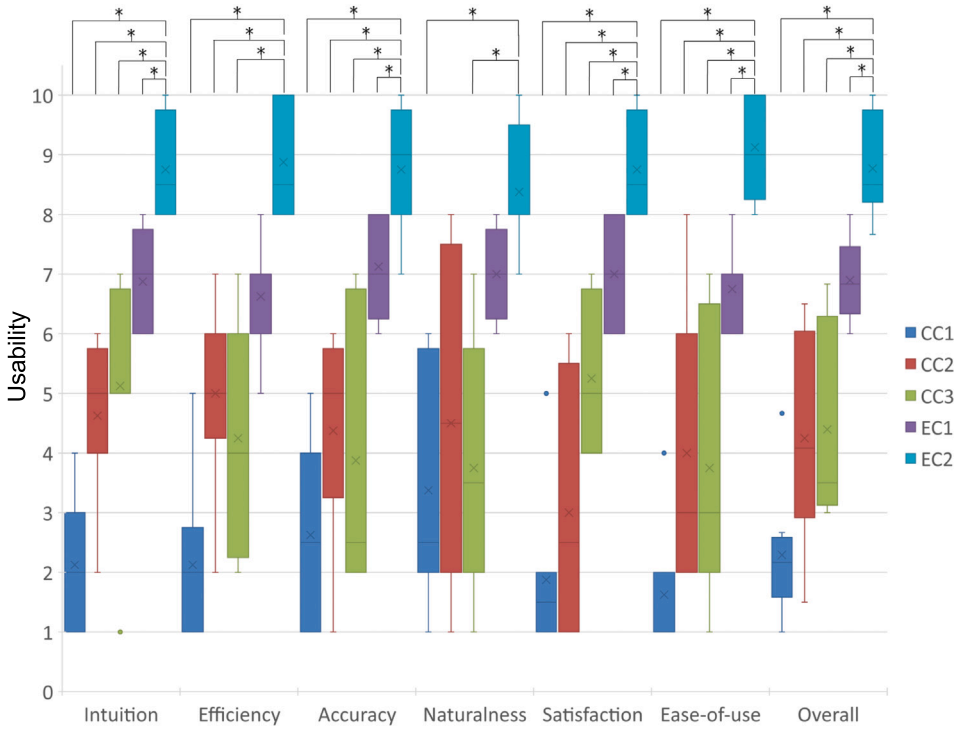


Fig. 10. Usability scores for six questions and overall. Significant differences are denoted with an asterisk.

Table 4
SSQ scores for different conditions.

Experimental condition	Avg ± std. dev.	Mean reduction	p	Cohen's d	Effect size
Before Task	0.82 ± 0.26	–	–	–	–
CC1	6.48 ± 1.05	85.9%	< 0.001*	2.70	Huge
CC2	4.79 ± 1.64	81.0%	< 0.001*	2.85	Huge
CC3	6.07 ± 1.51	85.0%	< 0.001*	4.05	Huge
EC1	1.82 ± 0.45	50.0%	0.118	0.89	Large
EC2	1.57 ± 0.14	–	–	–	–

showed that EC_2 was significantly ($p < 0.001$) improved in all six areas compared to CC_1 and CC_3. Compared to CC_2, EC_2 was significantly ($p < 0.001$) improved in intuition, efficiency, accuracy, satisfaction, and ease of use. Compared to EC_1, EC_2 significantly improved on intuition ($p = 0.001$), efficiency ($p < 0.001$), accuracy ($p = 0.004$), naturalness ($p = 0.01$), satisfaction ($p = 0.002$), and ease of use ($p < 0.001$). For the overall score, EC_2 significantly ($p < 0.001$) improved compared to CC_1, CC_2, CC_3, and EC_1. This result supports ***H6***.

Participants completed SSQ questionnaire before and after the task. The pre-task SSQ mean score was 0.82 with a standard deviation of 0.26, indicating that the participants' overall state was relatively stable and comfortable before the task. After completing the task, the participants' ratings for each condition showed significant differences. Specifically, the SSQ score for EC_1 was 1.82 ± 0.45, while the score for EC_2 was 1.57 ± 0.14. These two conditions had significantly lower scores compared to CC_1, CC_2, and CC_3, which were 6.48 ± 1.05, 4.79 ± 1.64, and 6.07 ± 1.51, respectively. Post hoc analyses showed that the SSQ score for EC_2 was significantly ($p < 0.001$) lower than CC_1, CC_2, and CC_3. Compared to EC_1, the reduction is not significant ($p = 0.118$). This result supports ***H7*** (Table 4).

4.7. Discussion

We discuss the advantages of our proposed user perception-based guided label layout approach over traditional dynamic label layout in terms of three key aspects: time efficiency, task load, and usability. Then, we also discuss the limitations of the user study.

Efficiency. As can be seen from Table 1, the average time to complete the task is significantly shorter in the EC_2 than in the CC_1, CC_2 and CC_3. This is because EC_2 significantly reduces the number of labels the user needs to view by limiting the number of labels compared to CC_1, and CC_2. Compared to CC_3, EC_2 greatly reduces unnecessary movement and rotation of the user's view by concentrating the labels of the objects that may interest the user within the user's view. Additionally, since the EC_2 method includes both improved layout and color coding, while the CC approaches do not, the observed performance improvement may come from these two factors or their interaction. Compared to EC_1, EC_2 is able to make a large improvement in reducing the time the user spends looking for objects. This is because EC_2 further reduces the user's time to find the object by optimizing the layout of the label positions in the view and arranging the labels in views with shorter user-perceived time based on the interest score. These results fully support the hypothesis **H1** and demonstrate the effectiveness of our approach in improving the efficiency of target localization.

Task load. As shown in Fig. 9, on all five questions of the NASA-TLX questionnaire, EC_2 scored significantly lower than CC_1, CC_2 and CC_3. This is because in the CC_1, CC_2 and CC_3 conditions, as shown in Tables 3 and 2, the user needs to keep moving and rotating the viewpoint in order to see the many objects and labels in the scene to locate the target, and thus the task load is higher. Furthermore, according to the results of the SSQ, some subjects experience more vertigo in the CC_1, CC_2, and CC_3. These results fully support hypothesis **H5** and demonstrate that our approach significantly reduces task load.

Usability. As shown in Fig. 10, EC_2 scores significantly higher than CC_1, CC_2, and CC_3 on all six questions except for naturalness in CC_2, which greatly improves the user experience. With fewer labels displayed per viewport, users experience a more intuitive, natural and easy-to-use interaction. As shown in Fig. 8, the perceived time cost of EC_2 is lower than that of CC_1, CC_2, CC_3, and EC_1, which improves the user's perception of its effectiveness and satisfaction. In addition, our

method is able to place labels of greater interest at locations with shorter perception times and is therefore more accurate. These results fully support hypothesis **H6** and show that our method has higher usability.

Limitation. A limitation of this study's experimental design is that color was not uniform across the experimental and control groups. While this prevents us from precisely quantifying the specific contribution of the layout algorithm to the performance improvement in EC_2 vs. CC, the EC_2 vs. EC_1 comparison and qualitative user feedback provide strong evidence. In user interviews, most users mentioned the "natural grouping of labels in EC_2" and "fewer occlusions" as key factors in helping them complete tasks faster. This suggests that improvements to the layout algorithm were a primary driver of performance improvement. Color likely played a secondary, enhancing role, and future research could use it as an independent variable in controlled experiments to verify its specific impact.

5. Conclusions, limitations and future work

We have proposed a user perception-based label layout method to enhance target localization efficiency. Our approach analyzes the user's level of interest in each object during the localization task and places labels at 3D spatial locations with shorter perceived times for objects that attract the user's attention. By optimizing the label layout, our method improves the efficiency of label-guided target localization. The results from user studies demonstrate that our label layout method significantly enhances target localization efficiency, reduces task load, and improves usability when compared to traditional dynamic label layouts.

Our approach is well-suited for the following example application scenario. In a pharmaceutical cleanroom, a technician must select a specific test vial among hundreds of physically or virtually identical vials. The digital twin of the laboratory contains metadata linking each vial to a batch ID stored in a database. However, the rendering system streams a live scene without object-level graphical access—meaning the application can retrieve the label (e.g., "Batch 001"), but cannot visually modify only the correct vial because the system lacks highlight hooks, semantic segmentation, or model-level access. In this case, conventional visual cueing (coloring, pulsing, outlining) is not possible, whereas the proposed semantic-search-based navigation allows the user to locate the target object efficiently.

However, there are several limitations to our approach. The first limitation is that our method is currently designed for static scenes, as the computation of the object interest score relies on pre-computed environmental similarity. The computation takes 13 minutes on average. In dynamic scenes, if an object moves, the environmental similarity between objects must be recalculated. If this is not taken into account, our method can be used in dynamic scenes. How to take into account the influence of the environment in which the object is located in dynamic scenes is part of our future work.

The second limitation is that when the target is not similar to other objects in the scene, the efficiency of target localization is comparable to that of traditional methods. This occurs because the interest score of the target does not increase when the user focuses their attention on other objects, which causes our method to place the target label in a less noticeable location, reducing its effectiveness. Another limitation is that the user studies were confounded by the inconsistent use of color across conditions. Although we demonstrated the effectiveness of the layout algorithm through internal comparisons, future work requires more rigorous controlled experiments to fully decouple the effects of layout and color on task performance.

Our approach focuses primarily on target localization in VR environments. However, if we were to extend this approach to AR scenes and apply it to the localization of real-world objects, we would need to detect user-visible objects in real-time and compute their similarity to other real objects (For now, it takes about 13 minutes on average to complete the computations). This real-time processing and environment similarity computation will be a key area for our future research.

CRediT authorship contribution statement

Jian Wu: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shuai Luan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dong Zhou:** Writing – review & editing, Conceptualization. **Wei Ke:** Writing – review & editing. **Lili Wang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered as potential competing interests:

Lili Wang reports financial support was provided by National Natural Science Foundation of China. Jian Wu reports financial support was provided by National Natural Science Foundation of China. Lili Wang reports financial support was provided by National Key Research and Development Program of China. Jian Wu reports financial support was provided by Fundamental Research Funds for the Central Universities. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.ijhcs.2026.103801.

Data availability

Data will be made available on request.

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